

E-COMMERCE SALES AND CUSTOMER INSIGHTS ANALYSIS

Data Visualization Final Project Report

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Tools Used: Microsoft Excel, Python, Pandas, Matplotlib, Power BI

Project Type: Data Visualization & Business Analytics

Academic / Training Project

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1. Introduction

Data visualization is the process of presenting data in a graphical and meaningful form so that patterns, trends, comparisons and business insights can be understood easily. This project focuses on an e-commerce dataset and converts raw transaction data into useful information through Excel, Python and Power BI.

The project analyzes sales, profit, orders, customers, product performance, categories and regional performance. The final outcome is an interactive dashboard and a set of visualizations that support quick business understanding and decision-making.

2. Project Objectives

- To understand and clean an e-commerce sales dataset.
- To calculate important business KPIs such as total sales, profit, orders and customers.
- To analyze sales and profit by category, region and product.
- To identify sales trends over time.
- To create visualizations using Python.
- To build an interactive dashboard using Power BI.
- To present data-driven insights in a simple and understandable way.

3. Problem Statement

Raw e-commerce transaction data contains a large number of records and is difficult to interpret directly. The objective is to transform this raw data into clear visual information that can answer questions such as: Which category generates the highest sales? Which region performs best? Which products are top performers? How do sales change over time? The project addresses these questions through data analysis and visualization.

4. Dataset Description

The project dataset contains 600 e-commerce order records covering the year 2025. It includes customer, order, product, geographic, pricing, discount, sales and profit information.

Column	Description
Order_ID	Unique order identifier
Order_Date	Date on which the order was placed
Customer_ID / Customer_Name	Customer information
Region / State	Geographical information
Category / Product	Product classification and product name
Unit_Price	Price per unit
Quantity	Number of units purchased
Discount	Discount applied to the order

Column	Description
Sales	Final sales value
Profit	Profit generated from the order
Month / Quarter	Time-based fields used for analysis

5. Tools and Technologies

Tool / Technology	Purpose
Microsoft Excel	Data checking, cleaning, Pivot Tables and basic analysis
Python	Data analysis and automation
Pandas	Data loading, cleaning, grouping and calculations
Matplotlib	Creating charts and visualizations
Power BI	Interactive dashboard and business reporting

6. Methodology

Step 1: Data Collection: The e-commerce transaction dataset was prepared for analysis.

Step 2: Data Cleaning: Duplicate records and missing values in important fields were checked and handled using Python.

Step 3: Data Analysis: Sales, profit, orders, customers and quantities were calculated.

Step 4: Exploratory Analysis: Data was grouped by month, category, region and product.

Step 5: Visualization: Charts were created using Python and Power BI.

Step 6: Dashboard Development: Important KPIs, charts and slicers were arranged into an interactive dashboard.

Step 7: Insight Generation: Major trends and high-performing areas were identified from the results.

7. Excel Analysis

Excel was used for initial data exploration and business analysis. The dataset was converted into an Excel Table and Pivot Tables were used to summarize sales and profit.

- Monthly Sales Trend — used to observe changes in sales over time.
- Sales by Category — used to compare category performance.
- Profit by Category — used to identify profitable categories.
- Sales by Region — used to compare geographical performance.
- Top 10 Products — used to identify products with the highest sales.
- Quantity Sold by Category — used to compare product volume.

8. Python Analysis and Visualization

Python was used to automate data analysis and create visualizations. Pandas was used for data manipulation and Matplotlib was used for graphical representation.

KPI	Result
Total Sales	■24,258,812.40
Total Profit	■6,153,941.29
Total Orders	600
Total Customers	177
Total Quantity	1,831
Average Order Value	■40,431.35
Profit Margin	25.37%

Python visualizations generated:

- Monthly Sales Trend
- Sales by Category
- Profit by Region
- Top 10 Products by Sales

9. Power BI Dashboard

Power BI was used to create the final interactive dashboard. The dashboard combines KPI cards, charts and slicers so that users can explore performance by region, category and quarter.

Dashboard Element	Purpose
Total Sales Card	Shows overall sales value
Total Profit Card	Shows overall profit
Total Orders Card	Shows number of unique orders
Total Customers Card	Shows number of unique customers
Monthly Sales Trend	Shows sales movement over time
Sales by Category	Compares category-level sales
Sales by Region	Compares regional sales
Top 10 Products	Highlights best-selling products
Profit by Category / Region	Shows profitability
Slicers	Filters dashboard by region, category and quarter

Screenshot Placement:

Insert a screenshot of the completed Power BI dashboard here before final submission. Recommended caption: *Figure 1: E-Commerce Sales and Customer Insights Dashboard.*

You can also insert the four Python chart screenshots after this section with captions such as *Figure 2: Monthly Sales Trend*, *Figure 3: Sales by Category*, *Figure 4: Profit by Region*, and *Figure 5: Top 10 Products by Sales*.

10. Key Findings and Insights

- The highest sales category in the analyzed dataset is Technology.
- The region with the highest total sales is West.
- The region with the highest total profit is West.
- The highest-sales product in the dataset is Headphones.
- The monthly sales visualization helps identify periods of higher and lower business activity.
- Category and regional comparisons help identify areas that can receive more attention in future business planning.

11. Learning Outcomes

- Learned how to work with structured business datasets.
- Improved understanding of data cleaning and preprocessing.
- Learned how to use Excel Pivot Tables for analysis.
- Practiced Python Pandas for data manipulation.
- Created business charts using Matplotlib.

- Learned the basics of Power BI dashboard development.
- Understood how visualizations can support business decision-making.
- Improved data interpretation and presentation skills.

12. Conclusion

The E-Commerce Sales and Customer Insights project successfully demonstrates the complete data visualization workflow, from raw data preparation and analysis to graphical presentation and dashboard creation. Excel provided a simple environment for data exploration, Python supported automated analysis and visualization, and Power BI provided an interactive way to communicate business insights. The project shows how data visualization can make complex transaction data easier to understand and use for decision-making.

13. Future Scope

- Add real-time sales data and automated dashboard refresh.
- Include customer segmentation and retention analysis.
- Add forecasting for future sales and profit.
- Analyze discount impact on profit.
- Add state-level maps and more detailed geographic analysis.
- Use advanced machine learning techniques for sales prediction.

Project End